AI Analytics Overview

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AI Analytics is one of GCXONE's most powerful features, using intelligent algorithms to analyze video footage and distinguish real security threats from false alarms. This technology can reduce false alarms by approximately 80%, significantly reducing the burden on security operators while ensuring critical alarms are processed within strict service level agreements.

Key Capabilities

AI-Powered False Alarm Filtering

Traditional motion detection systems generate numerous false alarms from animals, environmental factors such as wind and shadows, lighting changes, camera movement or vibration, and reflections and glare.

GCXONE's AI Analytics uses advanced computer vision algorithms to analyze video frames and identify actual security threats versus false triggers, reducing false alarms by up to 80%.

Detection Capabilities

Human Detection:

- Accurately identifies human figures in video footage
- Draws bounding boxes around detected persons
- Provides confidence levels for detections
- Tracks person movement across camera fields of view

Vehicle Detection:

- Identifies various vehicle types including cars, trucks, motorcycles, buses, bicycles, and trains
- Distinguishes between parked and moving vehicles
- Analyzes vehicle characteristics for classification

Object Detection:

- Identifies various objects that may trigger alarms
- Supports detection of specific custom object types
- Analyzes object behavior patterns

Centralized Alarm Processing

1. System receives alarms from connected devices
2. AI algorithms analyze associated video footage
3. Alarms classified as real (human or vehicle detected) or false (no threat)
4. Processed alarms distributed to operators based on customer preferences

5. Unclear cases forwarded for operator verification

Strict Service Level Agreements (SLA)

60–90 Second Processing Window: GCXONE is engineered to process alarms within a strict 60–90 second window, ensuring rapid response to genuine security threats.

Automatic Escalation: If the AI cannot process an alarm within the SLA timeframe, it automatically forwards it as a real alarm for human review — no genuine threat is missed.

Priority Classification: The system automatically classifies alarms based on detection confidence levels, threat type, alarm source and context, and historical patterns.

Alarm Quad Views

For simultaneous monitoring during incidents, GCXONE provides Alarm Quad Views displaying:

- Pre-Alarm Image: Video frame before the alarm event
- Current Image: Video frame at the moment of alarm
- Post-Alarm Image: Video frame after the alarm event
- AI Analysis Result: Bounding boxes and detection annotations

How It Works

1. Device generates alarm and sends to GCXONE
2. System captures video frames associated with alarm
3. AI algorithms analyze frames for object presence, characteristics, and scene context
4. Alarm classified as real or false based on analysis
5. System assigns confidence level to classification
6. Processed alarm routed to appropriate operator or workflow
7. Unclear or high-priority alarms sent for human review

Supported Alarm Types

- Motion Detection Alarms: Analyzes motion triggers for real vs. false
- Intrusion Detection: Verifies human or vehicle presence in restricted areas
- Line Crossing: Confirms object type crossing defined lines
- Loitering Detection: Identifies person vs. other objects loitering
- Tamper Detection: Distinguishes real tampering from environmental changes

Use Cases

Commercial Security

- Retail Stores: Filter motion alarms from customers vs. after-hours intrusions
- Office Buildings: Distinguish employees from unauthorized access
- Warehouses: Identify human vs. vehicle movement
- Parking Lots: Detect vehicles vs. animals or environmental triggers

Critical Infrastructure

- Perimeter Protection: Verify human presence at boundaries
- Access Control Points: Confirm vehicle types at entry points

Remote Monitoring

- Unmanned Sites: Essential for sites without on-site security
- After-Hours Monitoring: Critical for detecting real threats during closed hours
- Multi-Site Operations: Consistent filtering across multiple locations

Limitations and Considerations

Accuracy Factors

- Lower quality video may reduce detection accuracy
- Extreme lighting conditions — very dark or very bright — affect accuracy
- Optimal detection requires a clear camera view of the detection area
- Severe weather may impact analysis

Configuration Requirements

- Cameras must be positioned for optimal detection
- Proper configuration of detection areas improves accuracy
- Balancing sensitivity vs. false alarm rate requires threshold tuning

Best Practices

Camera Configuration

- Position cameras for a clear view of detection areas
- Ensure sufficient lighting for video analysis
- Maintain camera focus and secure mounting to prevent vibration

Detection Zone Setup

- Set detection zones to focus on relevant areas
- Exclude areas prone to false triggers such as trees and shadows
- Regularly test and fine-tune detection zones

Threshold Tuning

- Begin with higher confidence thresholds
- Track false alarm rates and adjust as needed
- Periodically review and adjust thresholds